On the Decidability of $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$: Quasimodels Meet the Patchwork Property

Claude*

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Abstract

The description logics $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$ extend $\mathcal{ALCI}$ with role boxes derived from the composition tables of the Region Connection Calculi RCC5 and RCC8. These logics were introduced by Wessel [10], who established PSPACE-hardness of $\mathcal{ALCI}_{RCC5}$ and EXPTIME-hardness of $\mathcal{ALCI}_{RCC8}$ but left decidability open. We settle both questions positively: concept satisfiability in $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$ is decidable, with an EXPTIME upper bound. The proof introduces a *quasimodel method* tailored to complete-graph models and exploits the *patchwork property* of the RCC calculi—the fact that path-consistent atomic constraint networks are globally consistent [8]—to lift local (triple-wise) consistency to global model existence via a Henkin construction. For $\mathcal{ALCI}_{RCC8}$ this yields EXPTIME-completeness.

Keywords: description logic, spatial reasoning, Region Connection Calculus, decidability, quasimodel, patchwork property

1 Introduction

Description logics (DLs) are a family of knowledge representation formalisms that underpin ontology languages such as OWL [1]. The *standard* decision procedures for DLs exploit the *tree model property*: every satisfiable concept has a tree-shaped model, so one can search for a finite abstraction of such a tree (a "type elimination" or "tableau" argument).

The $\mathcal{ALCI}_{RCC}$ family, introduced by Wessel [10, 9], takes a different path. Here the roles are not abstract binary relations but the *base relations* of a qualitative spatial calculus—specifically, a Region Connection Calculus (RCC) [6]. Interpretations are constrained to be complete graphs in which every pair of domain elements is related by exactly one base relation, and the composition of any two base relations along a path is required to conform to the RCC composition table.

Formally, let $\mathcal{T}$ be one of RCC1, RCC2, RCC3, RCC5, or RCC8. The logic $\mathcal{ALCI}_{RCC}[\mathcal{T}]$—or simply $\mathcal{ALCI}_{RCCk}$ for the standard RCC-k variants—is the extension of $\mathcal{ALCI}$ (the basic DL $\mathcal{ALC}$ enriched with inverse roles) whose role names are the base relations of $\mathcal{T}$, and whose interpretations are subject to the *one-cluster, JEPD, composition table, and converse constraints* (Section 2).

Wessel [10] classified most of the family:

- $\mathcal{ALCI}_{RCC1}$ is NP-complete (equivalent to the modal logic S5).
- $\mathcal{ALCI}_{RCC2}$ is decidable (reduction to a two-variable fragment).
- $\mathcal{ALCI}_{RCC3}$ is decidable (reduction to the two-variable fragment of first-order logic with equality, $\mathrm{FO}^2[=]$).

*Claude is an AI assistant by Anthropic. This research was prompted by Michael Wessel (miacwess@gmail.com), who introduced the $\mathcal{ALCI}_{RCC}$ family in his doctoral work at the University of Hamburg and left the decidability of $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$ as open problems [10, 9].

- $\mathcal{ALCI}_{RCC5}$ is PSPACE-hard; decidability was left open.
- $\mathcal{ALCI}_{RCC8}$ is EXPTIME-hard; decidability was left open.

The difficulty with $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$ stems from the combination of three properties: models are complete graphs (no tree model property), infinite models are required (no finite model property), and the composition tables have non-deterministic entries (the role boxes are not "admissible" in the sense of $\mathcal{ALCRA}_{SG}$ [9]).

Contribution. We prove that concept satisfiability in $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$ is decidable, with an EXPTIME upper bound. Combined with the known lower bounds, this yields EXPTIME-completeness for $\mathcal{ALCI}_{RCC8}$. For $\mathcal{ALCI}_{RCC5}$, the exact complexity remains between PSPACE and EXPTIME.

Proof technique. We develop a *quasimodel method* that abstracts a (possibly infinite) complete-graph model into a finite structure consisting of *types* (sets of subconcepts true at a node), *pair-types* (role-labeled type pairs), and *triple-types* (composition-consistent triples). The key step in the completeness direction—constructing a model from a quasimodel—uses the *patchwork property* of RCC5/RCC8 [8, 7]: path-consistent atomic constraint networks over these calculi are globally consistent. This allows us to extend a partial model by one element at a time, choosing roles to all existing elements simultaneously, with only pairwise (triple-wise) consistency checks needed.

Related work. Lutz and Wolter [5] studied modal logics with RCC8 topological modalities interpreted over actual topological spaces; in that setting, reasoning is generally undecidable. Lutz and Miličić [4] showed that $\mathcal{ALC}$ extended with ω-admissible concrete domains (which include RCC5 and RCC8) is decidable, but the concrete-domain approach does not allow quantification *over* spatial roles (the defining feature of $\mathcal{ALCI}_{RCC}$). Borgwardt et al. [2] further refined the complexity of concrete-domain reasoning. The $\mathcal{ALCRA}_{SG}$ fragment [9] handles *admissible* role boxes (deterministic, functional, complete, associative) via reduction to $\mathcal{ALC}$ with general TBoxes, but the RCC5/RCC8 role boxes are non-deterministic and hence fall outside $\mathcal{ALCRA}_{SG}$.

2 Preliminaries

2.1 The Description Logic $\mathcal{ALCI}_{RCC5}$

Definition 2.1 (Syntax)**.** Let N_C be a set of *concept names* and let $N_R = \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ be the set of *RCC5 base relations*, serving as role names. *Concepts* are built by the grammar:

$$C, D \ ::= \ A \mid \neg C \mid C \sqcap D \mid C \sqcup D \mid \exists R.C \mid \forall R.C \mid \exists R^{-}.C \mid \forall R^{-}.C$$

where $A \in N_C$ and $R \in N_R$. The *inverse* mapping on roles is $\mathsf{inv}(\mathsf{PP}) = \mathsf{PPI}$, $\mathsf{inv}(\mathsf{PPI}) = \mathsf{PP}$, $\mathsf{inv}(\mathsf{DR}) = \mathsf{DR}$, $\mathsf{inv}(\mathsf{PO}) = \mathsf{PO}$, $\mathsf{inv}(\mathsf{EQ}) = \mathsf{EQ}$.

Definition 2.2 (Semantics)**.** An $\mathcal{ALCI}_{RCC5}$ *interpretation* is a structure $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ where $\Delta^{\mathcal{I}} \neq \emptyset$ and $\cdot^{\mathcal{I}}$ assigns to each concept name A a set $A^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$ and to each base relation R a binary relation $R^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}}$, subject to:

(S1) **One cluster.** For all $x, y \in \Delta^{\mathcal{I}}$, there exists $R \in N_R$ with $(x, y) \in R^{\mathcal{I}}$.
(S2) **JEPD.** For distinct $R, S \in N_R$, $R^{\mathcal{I}} \cap S^{\mathcal{I}} = \emptyset$.
(S3) **Composition.** For all $R, S \in N_R$: $R^{\mathcal{I}} \circ S^{\mathcal{I}} \subseteq \bigcup_{T \in \mathsf{comp}(R,S)} T^{\mathcal{I}}$, where comp is the RCC5 composition table (Table 1).
(S4) **Converse.** $\mathsf{PPI}^{\mathcal{I}} = (\mathsf{PP}^{\mathcal{I}})^{-1}$, $\mathsf{DR}^{\mathcal{I}} = (\mathsf{DR}^{\mathcal{I}})^{-1}$, $\mathsf{PO}^{\mathcal{I}} = (\mathsf{PO}^{\mathcal{I}})^{-1}$.
(S5) **Strong EQ.** $\mathsf{EQ}^{\mathcal{I}} = \{(x, x) \mid x \in \Delta^{\mathcal{I}}\}$.

Concept constructors are interpreted standardly: $(\exists R.C)^{\mathcal{I}} = \{x \mid \exists y.\, (x, y) \in R^{\mathcal{I}} \wedge y \in C^{\mathcal{I}}\}$, etc.

A concept C is *satisfiable* if there exists an interpretation $\mathcal{I}$ and $x \in \Delta^{\mathcal{I}}$ with $x \in C^{\mathcal{I}}$.

Table 1: The RCC5 composition table. Entry for row S, column R gives $\mathsf{comp}(R, S)$, i.e., the possible relations for (a, c) when $R(a, b)$ and $S(b, c)$. The symbol $*$ denotes N_R, i.e., all five relations are possible.

comp	$\mathsf{DR}(a,b)$	$\mathsf{PO}(a,b)$	$\mathsf{EQ}(a,b)$	$\mathsf{PPI}(a,b)$	$\mathsf{PP}(a,b)$
$\mathsf{DR}(b,c)$	$*$	DR, PO, PPI	DR	DR, PO, PPI	DR
$\mathsf{PO}(b,c)$	DR, PO, PP	$*$	PO	PO, PPI	DR, PO, PP
$\mathsf{EQ}(b,c)$	DR	PO	EQ	PPI	PP
$\mathsf{PP}(b,c)$	DR, PO, PP	PO, PP	PP	PO, EQ, PP, PPI	PP
$\mathsf{PPI}(b,c)$	DR	DR, PO, PPI	PPI	PPI	$*$

Remark 2.3 (Weak EQ semantics)**.** The variant with *weak* EQ (only $Id_{\Delta^{\mathcal{I}}} \subseteq \mathsf{EQ}^{\mathcal{I}}$, so EQ may also hold between distinct elements) reduces to the strong variant by adding the axiom $\top \sqsubseteq \forall \mathsf{EQ}.C \sqcap \forall \mathsf{EQ}^{-}.C$ for every relevant concept C and internalizing via the universal role $R_* = \mathsf{DR} \sqcup \mathsf{PO} \sqcup \mathsf{EQ} \sqcup \mathsf{PP} \sqcup \mathsf{PPI}$. See Wessel [10, Section 4.3]. We therefore work exclusively with strong EQ.

Remark 2.4 (Key structural properties)**.** In any $\mathcal{ALCI}_{RCC5}$ interpretation:

- PP is a strict partial order (irreflexive, asymmetric, transitive).
- There is no finite model property: the concept $(\exists \mathsf{PP}.\top) \sqcap (\forall \mathsf{PP}.\exists \mathsf{PP}.\top)$ is satisfiable only in infinite models.
- There is no tree model property: all models are complete graphs.
- The *universal role* $R_* = \mathsf{DR} \sqcup \mathsf{PO} \sqcup \mathsf{EQ} \sqcup \mathsf{PP} \sqcup \mathsf{PPI}$ is definable, enabling TBox internalization.
- Under strong EQ, a *difference role* $R_D = R_* \setminus \mathsf{EQ}$ is available, enabling limited nominal expressivity.

2.2 The Patchwork Property of RCC5

We recall the fundamental result from qualitative constraint reasoning that is the lynchpin of our decidability proof.

Definition 2.5 (Atomic constraint network)**.** An *atomic RCC5 constraint network* N over a set of variables V assigns to each ordered pair (v_i, v_j) with $v_i \neq v_j$ a single base relation $R_{ij} \in N_R \setminus \{\mathsf{EQ}\}$, and assigns EQ to each (v_i, v_i), respecting converse ($R_{ji} = \mathsf{inv}(R_{ij})$).

Definition 2.6 (Path consistency)**.** An atomic network N is *path-consistent* if for every triple of variables (v_i, v_j, v_k):

$$R_{ik} \in \mathsf{comp}(R_{ij}, R_{jk}).$$

Definition 2.7 (Consistency)**.** An atomic network N is *consistent* (or *realizable*) if there exists an $\mathcal{ALCI}_{RCC5}$ interpretation $\mathcal{I}$ and elements $d_v \in \Delta^{\mathcal{I}}$ for each $v \in V$ such that $(d_{v_i}, d_{v_j}) \in R_{ij}^{\mathcal{I}}$ for all i, j.

Theorem 2.8 (Renz & Nebel [8]; Renz [7])**.** *For RCC5 (and for RCC8), an atomic constraint network is consistent if and only if it is path-consistent.*

Corollary 2.9 (Patchwork property)**.** *Let N_1 and N_2 be consistent atomic RCC5 networks that agree on their common variables. Then $N_1 \cup N_2$ is consistent.*

Proof. Path consistency is a local (triple-wise) condition. Every triple in $N_1 \cup N_2$ either lies entirely in N_1, entirely in N_2, or involves variables from both—in the last case, the shared variables carry the same labels in both networks, and path consistency of the triple follows from the individual path consistencies and the agreement on shared variables. By Theorem 2.8, $N_1 \cup N_2$ is consistent. □

The patchwork property is the engine of our decidability proof: it ensures that *local (triple-wise) consistency implies global consistency* for atomic RCC5 networks.

3 Monotonicity Along PP-Chains

Before developing the quasimodel method, we establish a structural property of $\mathcal{ALCI}_{RCC5}$ models that constrains the behavior of infinite PP-chains and explains why $\mathcal{ALCI}_{RCC5}$ is "tamer" than one might expect.

Lemma 3.1 (Upward monotonicity)**.** *Let $\mathcal{I}$ be an $\mathcal{ALCI}_{RCC5}$ interpretation containing a PP-chain $a_0, a_1, a_2, \ldots$ (i.e., $(a_i, a_{i+1}) \in \mathsf{PP}^{\mathcal{I}}$ for all i). For any $x \in \Delta^{\mathcal{I}}$ and any $i \geq 0$, the relation $R(x, a_{i+1})$ is constrained by $R(x, a_i)$ as follows:*

$R(x, a_i)$	*Possible values of* $R(x, a_{i+1})$
PP	PP
PO	PO, PP
DR	DR, PO, PP
PPI	PO, EQ, PP, PPI

Proof. Direct from the composition table: $R(x, a_{i+1}) \in \mathsf{comp}(R(x, a_i), \mathsf{PP})$. Reading column $\mathsf{PP}(a, b)$ in Table 1 for each row gives the claimed sets. □

Corollary 3.2. *Going upward along a PP-chain, the relation of an external node progresses monotonically through at most four phases. The "strength" ordering* DR < PO < PP *is respected (with* PPI *on a separate track that merges into* PO/PP*). Reversals* PP → PO*,* PO → DR*, or* PP → DR *are impossible.*

Lemma 3.3 (Downward persistence)**.** *Going* downward *along a PP-chain (i.e., from a_i to a_{i-1} via* PPI*), the relation $R(x, a_{i-1}) \in \mathsf{comp}(R(x, a_i), \mathsf{PPI})$ satisfies:*

$R(x, a_i)$	*Possible values of* $R(x, a_{i-1})$
DR	DR
PPI	PPI
PO	PO, PPI
PP	PO, EQ, PP, PPI

In particular, DR *and* PPI *are downward closed.*

Proof. By $R(x, a_{i-1}) \in \mathsf{comp}(R(x, a_i), \mathsf{PPI})$, reading column $\mathsf{PPI}(a, b)$ from Table 1. □

Remark 3.4. The monotonicity lemmas show that an infinite PP-chain can only "absorb" external nodes in a structured way: once a node becomes DR or PPI relative to a chain element, it stays that way for all elements below; once it becomes PP, it stays PP for all elements above. This regularity is what allows a finite quasimodel to faithfully represent the infinite model structure.

4 The Quasimodel Method

Throughout this section, fix a concept C_0 in negation normal form (NNF).

4.1 Fischer–Ladner Closure

Definition 4.1 (Closure)**.** The *closure* $\mathsf{cl}(C_0)$ is the smallest set containing $\mathsf{sub}(C_0)$ (all subconcepts of C_0) and, for every concept name $A \in N_C$ occurring in C_0, the concept $\neg A$. We identify $\neg\neg D$ with D. Note $|\mathsf{cl}(C_0)| \leq 2|C_0|$.

4.2 Types

Definition 4.2 (Type)**.** A *type* (for C_0) is a subset $\tau \subseteq \mathsf{cl}(C_0)$ satisfying:
(T1) If $D_1 \sqcap D_2 \in \tau$, then $D_1 \in \tau$ and $D_2 \in \tau$.
(T2) If $D_1 \sqcup D_2 \in \tau$, then $D_1 \in \tau$ or $D_2 \in \tau$.
(T3) For every concept name A: not both $A \in \tau$ and $\neg A \in \tau$.
We write $\mathsf{Tp}(C_0)$ for the set of all types. Clearly $|\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|} \leq 2^{2|C_0|}$.

Given an interpretation $\mathcal{I}$ and $x \in \Delta^{\mathcal{I}}$, the type of x is $\tau^{\mathcal{I}}(x) = \{D \in \mathsf{cl}(C_0) \mid x \in D^{\mathcal{I}}\}$.

4.3 Pair-Types (R-Compatibility)

Definition 4.3 (R-compatibility)**.** Let $\tau_1, \tau_2 \in \mathsf{Tp}(C_0)$ and $R \in N_R \setminus \{\mathsf{EQ}\}$. The triple (τ_1, R, τ_2) is *R-compatible* if:
(P1) For all $\forall R.D \in \mathsf{cl}(C_0)$: if $\forall R.D \in \tau_1$ then $D \in \tau_2$.
(P2) For all $\forall \mathsf{inv}(R).D \in \mathsf{cl}(C_0)$: if $\forall \mathsf{inv}(R).D \in \tau_2$ then $D \in \tau_1$.
(P3) For all $\forall S.D \in \mathsf{cl}(C_0)$ with $S \neq R$ and $S \neq \mathsf{inv}(R)$: no new constraint is imposed by S on this pair. (The pair is related by R, not S.)
We define:

$$\mathsf{Pair}(C_0) = \{(\tau_1, R, \tau_2) \mid \tau_1, \tau_2 \in \mathsf{Tp}(C_0),\ R \in N_R \setminus \{\mathsf{EQ}\},\ (\tau_1, R, \tau_2) \text{ is } R\text{-compatible}\}.$$

Note that since we work with strong EQ semantics, the EQ relation holds only reflexively, and we handle it separately: $(\tau, \mathsf{EQ}, \tau)$ is always implicitly present for any type τ.

Remark 4.4. R-compatibility captures the *local* constraints imposed by universal restrictions. If $(\tau_1, R, \tau_2) \in \mathsf{Pair}(C_0)$, then two domain elements of types τ_1 and τ_2 could consistently be R-related—as far as universal restrictions are concerned.

4.4 Triple Consistency

Definition 4.5 (Consistent triple)**.** A *triple* $(\tau_1, R_{12}, \tau_2, R_{13}, \tau_3, R_{23})$ is *consistent* if:
(Tr1) Each pair is compatible: (τ_1, R_{12}, τ_2), (τ_1, R_{13}, τ_3), (τ_2, R_{23}, τ_3) are all in $\mathsf{Pair}(C_0)$.
(Tr2) The composition table is respected:

$$\begin{aligned} R_{13} &\in \mathsf{comp}(R_{12}, R_{23}), \\ R_{23} &\in \mathsf{comp}(\mathsf{inv}(R_{12}), R_{13}), \\ R_{12} &\in \mathsf{comp}(R_{13}, \mathsf{inv}(R_{23})). \end{aligned}$$

We write $\mathsf{Trip}(C_0)$ for the set of all consistent triples.

4.5 Quasimodel

Definition 4.6 (Quasimodel)**.** A *quasimodel* for C_0 is a triple $\mathcal{Q} = (T, P, \tau_0)$ where:
- $T \subseteq \mathsf{Tp}(C_0)$ is a non-empty set of types,
- $P \subseteq \mathsf{Pair}(C_0)$ is a set of pair-types restricted to T,
- $\tau_0 \in T$ is a distinguished type with $C_0 \in \tau_0$,

satisfying the following conditions:

(Q1) Existential demand satisfaction. For every $\tau \in T$ and every $\exists R.D \in \tau$, there exists a *witness type* $\tau' \in T$ with $D \in \tau'$ and $(\tau, R, \tau') \in P$, such that:
(Q1.a) *Pairwise embeddability with existing types:* For every $\tau'' \in T$ and every $R'' \in N_R \setminus \{\mathsf{EQ}\}$ with $(\tau, R'', \tau'') \in P$, there exists $S \in N_R \setminus \{\mathsf{EQ}\}$ with $(\tau'', S, \tau') \in P$ and $(\tau, R'', \tau'', R, \tau', S) \in \mathsf{Trip}(C_0)$.

(Q1.b) *Full triple coherence:* For every $\tau_a, \tau_b \in T$ and every R_{ab} with $(\tau_a, R_{ab}, \tau_b) \in P$, there exist $S_a, S_b \in N_R \backslash \{\mathsf{EQ}\}$ with $(\tau_a, S_a, \tau') \in P$ and $(\tau_b, S_b, \tau') \in P$ such that $(\tau_a, R_{ab}, \tau_b, S_a, \tau', S_b) \in \mathsf{Trip}(C_0)$.

(Q2) Completeness. For every $\tau_1, \tau_2 \in T$, there exists $R \in N_R \setminus \{\mathsf{EQ}\}$ with $(\tau_1, R, \tau_2) \in P$.

(Q3) PP-transitivity. If $(\tau_1, \mathsf{PP}, \tau_2) \in P$ and $(\tau_2, \mathsf{PP}, \tau_3) \in P$, then $(\tau_1, \mathsf{PP}, \tau_3) \in P$, and for all $\forall \mathsf{PP}.D \in \mathsf{cl}(C_0)$: if $\forall \mathsf{PP}.D \in \tau_1$ then $D \in \tau_3$. Symmetrically for PPI.

Remark 4.7. Condition (Q1) is the heart of the quasimodel. It ensures that when we add a new element (to witness an existential demand), we can assign it a role to *every* existing element in a way that is simultaneously consistent. Conditions (Q1.a) and (Q1.b) correspond to different perspectives on this requirement. Condition (Q2) ensures the one-cluster property. Condition (Q3) handles the transitivity of PP (and PPI), which propagates universal restrictions along PP-chains.

5 Main Theorem

Theorem 5.1. *A concept C_0 is satisfiable in $\mathcal{ALCI}_{RCC5}$ if and only if there exists a quasimodel for C_0.*

The proof occupies the next two subsections.

5.1 Soundness: Model ⇒ Quasimodel

Proof of soundness. Let $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ be an $\mathcal{ALCI}_{RCC5}$ interpretation with $x_0 \in C_0^{\mathcal{I}}$. Define:

$$\begin{aligned} T &= \{\tau^{\mathcal{I}}(x) \mid x \in \Delta^{\mathcal{I}}\}, \\ P &= \{(\tau^{\mathcal{I}}(x), R, \tau^{\mathcal{I}}(y)) \mid (x, y) \in R^{\mathcal{I}},\ R \in N_R \setminus \{\mathsf{EQ}\}\}, \\ \tau_0 &= \tau^{\mathcal{I}}(x_0). \end{aligned}$$

We verify each quasimodel condition:

(Q1) Let $\tau = \tau^{\mathcal{I}}(x)$ and $\exists R.D \in \tau$. Then there exists y with $(x, y) \in R^{\mathcal{I}}$ and $y \in D^{\mathcal{I}}$. Set $\tau' = \tau^{\mathcal{I}}(y)$.

For (Q1.a): given any $\tau'' = \tau^{\mathcal{I}}(z)$ with $(\tau, R'', \tau'') \in P$, witnessed by some z' with $\tau^{\mathcal{I}}(z') = \tau''$ and $(x, z') \in (R'')^{\mathcal{I}}$, the relation S between z' and y in $\mathcal{I}$ satisfies all required conditions by the model properties.

For (Q1.b): given any $\tau_a = \tau^{\mathcal{I}}(a)$, $\tau_b = \tau^{\mathcal{I}}(b)$ with $(\tau_a, R_{ab}, \tau_b) \in P$, witnessed by some a', b' with the appropriate types and relations, the relations of a' and b' to y provide the required S_a, S_b.

(Q2) By the one-cluster property, every pair of elements has a role, so every pair of types has a role in P.

(Q3) Transitivity of $\mathsf{PP}^{\mathcal{I}}$ ensures that if $(\tau_1, \mathsf{PP}, \tau_2) \in P$ and $(\tau_2, \mathsf{PP}, \tau_3) \in P$, then there exist elements x_1, x_2, x_3 such that $(x_1, x_2) \in \mathsf{PP}^{\mathcal{I}}$, $(x_2, x_3) \in \mathsf{PP}^{\mathcal{I}}$, and hence $(x_1, x_3) \in \mathsf{PP}^{\mathcal{I}}$, giving $(\tau_1, \mathsf{PP}, \tau_3) \in P$. The propagation of universal restrictions follows from the model satisfying $\forall \mathsf{PP}.D$. □

5.2 Completeness: Quasimodel ⇒ Model

This is the non-trivial direction.

Proof of completeness. Let $\mathcal{Q} = (T, P, \tau_0)$ be a quasimodel for C_0. We construct a model $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ via a *Henkin-style step-by-step construction.*

Stage 0. Create a single element e_1 with type $\tau(e_1) := \tau_0$. Set $\Delta_0 = \{e_1\}$.

Stage $n+1$. Let $\Delta_n = \{e_1, \ldots, e_m\}$ with types $\tau(e_i)$ and roles $R(e_i, e_j) \in N_R$ assigned for all $i \neq j$ (with $R(e_i, e_i) = \mathsf{EQ}$). Assume inductively:

(i) $(\tau(e_i), R(e_i, e_j), \tau(e_j)) \in P$ for all $i \neq j$.

(ii) Every triple (e_i, e_j, e_k) satisfies the composition table.

Enumerate all unsatisfied demands: pairs $(e_k, \exists R.D)$ where $\exists R.D \in \tau(e_k)$ but no element in Δ_n serves as a witness. Consider the next unsatisfied demand $(e_k, \exists R.D)$.

By condition (Q1) of the quasimodel, there exists a witness type $\tau' \in T$ with $D \in \tau'$ and $(\tau(e_k), R, \tau') \in P$. Create a new element e_{m+1} with $\tau(e_{m+1}) := \tau'$. Set $R(e_k, e_{m+1}) := R$ and $R(e_{m+1}, e_k) := \mathsf{inv}(R)$.

The key step is assigning a role $R(e_i, e_{m+1})$ for every existing element $e_i \in \Delta_n$ with $i \neq k$.

Claim 5.2. *There exist roles $S_1, \ldots, S_m \in N_R \setminus \{\mathsf{EQ}\}$ (with $S_k = R$) such that:*

(a) $(\tau(e_i), S_i, \tau') \in P$ for all i.

(b) For all $i \neq j$: the triple (e_i, e_j, e_{m+1}) with roles $R(e_i, e_j)$, S_i, S_j satisfies $S_i \in \mathsf{comp}(R(e_i, e_j), S_j)$ and is in $\mathsf{Trip}(C_0)$.

Proof of Claim 5.2. Consider the problem of choosing roles $S_1, \ldots, S_m$ as a constraint satisfaction problem. Define an atomic RCC5 constraint network N on the variables $\{e_1, \ldots, e_m, e_{m+1}\}$:

- For $i, j \leq m$: $N(e_i, e_j) = R(e_i, e_j)$ (the existing, consistent network).
- For $i \leq m$: $N(e_i, e_{m+1}) = S_i$ (to be determined), with $S_k = R$ fixed.
- The domain of S_i is $\{S \in N_R \setminus \{\mathsf{EQ}\} \mid (\tau(e_i), S, \tau') \in P\}$.

The constraints on the S_i are: for all $i \neq j$ with $i, j \leq m$,

$$S_i \in \mathsf{comp}(R(e_i, e_j), S_j) \quad \text{and} \quad S_j \in \mathsf{comp}(R(e_j, e_i), S_i).$$

Additionally, the triple $(\tau(e_i), R(e_i, e_j), \tau(e_j), S_i, \tau', S_j)$ must be in $\mathsf{Trip}(C_0)$.

Now, condition (Q1.b) of the quasimodel guarantees that for every *pair* (e_i, e_j), there *exist* values S_i, S_j satisfying these constraints. In other words, every pairwise sub-network $\{e_i, e_j, e_{m+1}\}$ is path-consistent.

We need these pairwise-consistent choices to extend to a *globally* consistent assignment. This is precisely the content of the **patchwork property** (Corollary 2.9).

More explicitly: the constraint network N restricted to any triple $\{e_i, e_j, e_{m+1}\}$ is consistent (by the pairwise argument above), so the full network N is path-consistent. By Theorem 2.8, a path-consistent atomic RCC5 network is globally consistent. Therefore, a simultaneous assignment $S_1, \ldots, S_m$ satisfying all constraints exists. □

Set $R(e_i, e_{m+1}) := S_i$ and $R(e_{m+1}, e_i) := \mathsf{inv}(S_i)$ for all i. This extends the interpretation to $\Delta_{n+1} = \Delta_n \cup \{e_{m+1}\}$ while maintaining the inductive invariants (i) and (ii).

The limit model. Using a fair enumeration of demands (e.g., dovetailing), define the limit interpretation:

$$\mathcal{I} = \bigcup_{n \geq 0} (\Delta_n, \cdot_n^{\mathcal{I}}).$$

This is a valid $\mathcal{ALCI}_{RCC5}$ interpretation:

- *One cluster:* every pair of elements created in the construction has a role assigned.
- *JEPD:* each pair has exactly one base relation (by construction).
- *Composition:* every triple satisfies the composition table (maintained at each stage by the claim).
- *Converse:* maintained by setting $R(e_{m+1}, e_i) = \mathsf{inv}(S_i)$.
- *Strong EQ:* $R(e_i, e_i) = \mathsf{EQ}$ for all i, and no distinct pair has EQ.
- *Concept truth:* for each type τ, every element e with $\tau(e) = \tau$ satisfies exactly the concept descriptions in τ. Since $C_0 \in \tau_0 = \tau(e_1)$, we have $e_1 \in C_0^{\mathcal{I}}$.

More precisely, concept truth is established by induction on concept structure:

- Concept names and negations: by the type conditions (T1)–(T3).
- Conjunctions, disjunctions: by (T1), (T2).
- Existential restrictions $\exists R.D$: every demand $\exists R.D \in \tau(e)$ is eventually satisfied by the Henkin construction (fair enumeration).
- Universal restrictions $\forall R.D$: if $\forall R.D \in \tau(e)$ and $(e, e') \in R^{\mathcal{I}}$, then $D \in \tau(e')$ by the R-compatibility condition (P1) in $\mathsf{Pair}(C_0)$.

Therefore $\mathcal{I} \models C_0$, and C_0 is satisfiable. □

6 Decidability and Complexity

Theorem 6.1. *Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable in* ExpTime.

Proof. By Theorem 5.1, it suffices to decide quasimodel existence. We describe a type-elimination algorithm.

1. **Initialization.** Compute $\mathsf{Tp}(C_0)$, $\mathsf{Pair}(C_0)$, and $\mathsf{Trip}(C_0)$. These sets have sizes bounded by $2^{O(|C_0|)}$, $2^{O(|C_0|)}$, and $2^{O(|C_0|)}$ respectively.
2. **PP-transitivity closure.** Close $\mathsf{Pair}(C_0)$ under condition (Q3): iteratively add $(\tau_1, \mathsf{PP}, \tau_3)$ whenever $(\tau_1, \mathsf{PP}, \tau_2)$ and $(\tau_2, \mathsf{PP}, \tau_3)$ are present and the universal restriction propagation is satisfied. Remove triples that violate (Q3).
3. **Type elimination.** Iteratively remove from T (initialized to $\mathsf{Tp}(C_0)$) any type τ that fails one of:
   - *Existential demand:* there exists $\exists R.D \in \tau$ for which no witness τ' satisfying (Q1) can be found.
   - *Completeness:* there exists $\tau' \in T$ such that no role R yields $(\tau, R, \tau') \in P$.

   When a type is removed, also remove all pair-types and triple-types involving it, then repeat.
4. **Termination.** Each iteration removes at least one type or pair-type. Since there are at most $2^{O(|C_0|)}$ types and $2^{O(|C_0|)}$ pair-types, termination occurs after at most $2^{O(|C_0|)}$ iterations.
5. **Acceptance.** Accept if and only if there remains a type τ_0 with $C_0 \in \tau_0$ such that τ_0 and the surviving types and pair-types form a quasimodel.

Each iteration inspects at most $2^{O(|C_0|)}$ types, pairs, and triples. Total time: $2^{O(|C_0|)} \cdot 2^{O(|C_0|)} = 2^{O(|C_0|)}$, which is in ExpTime. □

Theorem 6.2 (Lower bound)**.** *Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is PSpace-hard.*

Proof. Wessel [10, Section 6.3] establishes this by a reduction from QBF validity, using the PO-free fragment to enforce tree-shaped models. □

Corollary 6.3. *Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is between PSpace and ExpTime.*

Remark 6.4 (Closing the gap)**.** We conjecture that $\mathcal{ALCI}_{RCC5}$ is ExpTime-complete. The interaction between PP-transitivity and the universal role (which enables TBox internalization) plausibly supports a reduction from the ExpTime-hard *two-player corridor tiling problem*, but we have not established this.

7 Extension to $\mathcal{ALCI}_{RCC8}$

The RCC8 calculus uses eight base relations: DC, EC, PO, EQ, TPP, TPPI, NTPP, NTPPI. The logic $\mathcal{ALCI}_{RCC8}$ is defined analogously to $\mathcal{ALCI}_{RCC5}$, using the RCC8 composition table.

Theorem 7.1. *Concept satisfiability in $\mathcal{ALCI}_{RCC8}$ is decidable in* ExpTime.

Proof. The proof follows the same structure as for $\mathcal{ALCI}_{RCC5}$. The crucial ingredient is that RCC8 also enjoys the patchwork property: path-consistent atomic RCC8 networks are globally consistent (Renz [7]). Therefore:

- Types, pair-types, and triple-types are defined over the RCC8 base relations.
- The quasimodel conditions are stated with respect to the RCC8 composition table.
- The Henkin construction uses the RCC8 patchwork property at the key step (Claim 5.2).
- The type-elimination algorithm runs in $2^{O(|C_0|)}$ time.

The soundness and completeness arguments carry over verbatim. □

Theorem 7.2 (Wessel [10])**.** *Concept satisfiability in $\mathcal{ALCI}_{RCC8}$ is ExpTime-hard.*

Proof. By reduction from the two-player corridor tiling game [10, Section 6.4]. □

Corollary 7.3. *Concept satisfiability in $\mathcal{ALCI}_{RCC8}$ is ExpTime-complete.*

8 Related Work

Wessel's $\mathcal{ALCRA}_{SG}$ (2000/2001). The $\mathcal{ALCRA}_{SG}$ framework [9] handles role boxes that are *admissible*: deterministic (each composition yields a single relation), functional, complete, and associative. For such role boxes, the logic reduces to $\mathcal{ALC}$ with general TBoxes, yielding ExpTime-completeness. The RCC5 and RCC8 role boxes are non-deterministic (many composition entries yield sets of relations) and hence fall outside $\mathcal{ALCRA}_{SG}$. Our quasimodel method handles this non-determinism by working with all possible completions simultaneously, guided by the patchwork property.

Lutz & Wolter (2006): Modal logics of topological relations. Lutz and Wolter [5] study modal logics with RCC8 modalities interpreted over *actual topological spaces*. In their setting, models must correspond to genuine spatial configurations (e.g., in $\mathbb{R}^n$), which imposes additional constraints beyond the composition table. This leads to undecidability in many cases. By contrast, $\mathcal{ALCI}_{RCC}$ uses *abstract* semantics: any interpretation satisfying the composition table constraints is admissible. The abstract semantics are more permissive, and this permissiveness is exactly what makes our decidability proof work.

Lutz & Miličić (2007): ω-admissible concrete domains. Lutz and Miličić [4] prove that $\mathcal{ALC}$ extended with ω-admissible concrete domains is decidable. RCC5 and RCC8 are ω-admissible, so $\mathcal{ALC}$(RCC5) and $\mathcal{ALC}$(RCC8) are decidable. However, the concrete-domain approach is fundamentally different: spatial predicates constrain *concrete features* of domain elements via functional roles, and one cannot quantify *over* spatial relations. In particular, $\forall \mathsf{PP}.C$ and $\exists \mathsf{DR}.D$ are not expressible in $\mathcal{ALC}$(RCC5). The $\mathcal{ALCI}_{RCC}$ logics are strictly more expressive for spatial reasoning because spatial relations serve directly as roles.

Borgwardt, De Bortoli & Koopmann (2024). Borgwardt et al. [2] sharpen the complexity of reasoning in $\mathcal{ALC}$ with ω-admissible concrete domains. Their results apply to the concrete-domain setting and do not directly bear on $\mathcal{ALCI}_{RCC}$, but confirm that the patchwork/amalgamation property of RCC calculi is a powerful tool for decidability in description logics with spatial features.

Constraint satisfaction and qualitative reasoning. The patchwork property (Theorem 2.8) was established by Renz and Nebel [8] and Renz [7] in the context of qualitative spatial reasoning and constraint satisfaction. Our contribution is to bridge the gap between this CSP result and the description logic setting: we show that the patchwork property, when combined with the quasimodel method, is precisely the tool needed to decide satisfiability in $\mathcal{ALCI}_{RCC}$ logics.

9 Discussion and Open Questions

1. **Exact complexity of $\mathcal{ALCI}_{RCC5}$.** We have established PSPACE $\leq \mathcal{ALCI}_{RCC5} \leq$ EXPTIME. The PSPACE lower bound [10] uses only the PO-free fragment. We conjecture that the full logic is EXPTIME-complete, but closing this gap requires a new lower-bound argument.

2. **General characterization.** Which composition tables (beyond RCC5 and RCC8) yield decidable $\mathcal{ALCI}_{RCC}$-style logics? Our proof shows that the patchwork property is sufficient. It would be valuable to characterize exactly which qualitative calculi have this property and whether it is also necessary for decidability.

3. **Adding number restrictions.** Unqualified number restrictions make $\mathcal{ALCRA}_{SG}$ undecidable [9]. We expect the same for $\mathcal{ALCI}_{RCC5}$: number restrictions would allow enforcing specific branching patterns that, combined with concept distinctions, may enable grid encodings.

4. **Finite model reasoning.** Satisfiability restricted to *finite* models (ruling out infinite PP-chains) may be decidable in a lower complexity class, since the need for infinite witnesses—and hence the full power of the Henkin construction—disappears.

5. **Hybrid extensions.** Wessel [10] considers hybrid logic extensions with nominals and the @-operator. Whether our decidability results extend to these enrichments is an open question; the interaction between nominals and the complete-graph semantics may introduce new complications.

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